Sponsor Exposure Analytics for the Rest of Us: An Open, Self-Hostable Framework for Small and Mid-Tier Sports Clubs

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***Abstract*—Measuring how long and how prominently a sponsor's logo appears in match video, and converting that exposure into an estimated media value (EMV), is a well-served problem for clubs that can afford an enterprise vendor (Nielsen Sports, GumGum/Zoomph, Relo Metrics, Shikenso) or that maintain an in-house data-science team — and, in practice, for clubs whose matches are covered by an official broadcast feed those vendors can integrate with in the first place. Most professional clubs outside the top-revenue leagues have neither the budget nor a broadcast deal. We present an end-to-end, self-hosted framework that a club can use to ingest *its own* match video — whatever it is able to capture itself, not necessarily an official broadcast feed — and turn it into sponsor exposure and media-value reports. The framework is assembled entirely from open-source models and libraries, deployable on commodity hardware, and deployed and evaluated at a real professional sports club whose identity we withhold for confidentiality. Beyond detection, the framework resolves two problems that prior published systems leave open: attributing each detected logo to the correct team when the same sponsor appears on both sides' kits, advertising boards, or officials; and pricing sponsor placements at the level of individual kit slots rather than only aggregate screen time. We position our pricing formula explicitly against existing valuation practice — both the manual Advertising/Earned Media Value Equivalency (AVE) methodology used across the sponsorship industry, and recent AI-driven valuation systems — and show that neither line of work prices exposure below the level of a brand's total on-screen time. On self-ingested match video from a real fixture, the framework tracks players, filters out opponent-owned instances of shared sponsors, and produces per-brand and per-placement exposure and EMV reports; we report detection accuracy on a clip-disjoint split (mAP@0.5 = 0.75; the same detector evaluated under a naive random-frame split reaches 0.86, a gap we report explicitly as evidence of split-induced inflation rather than model quality), team-attribution behaviour on an initial set of labelled clips, and the scale of the overcounting that attribution prevents: across eight of nine full-match deployment videos, 41% of raw logo detections belonged to opponents or officials, or were unattached to any tracked player, and would otherwise have been priced as the target club's kit exposure; the ninth match, flagged by a visual audit, is reported as a failure case of the team classifier rather than as filtering. We report the design choices, calibration lessons, and failure modes needed to take a sponsor-analytics system from paper to a low-cost, production deployment.**

***Index Terms*—sponsor visibility, media value equivalency, logo detection, team attribution, sports analytics, open-source, accessibility, self-service video ingestion.**

I. Introduction

Sports clubs sell advertising space on their kits, but only a small number of clubs can verify what that space is actually worth. Our partner club — a professional club that sells sponsor slots on its kit but, like most clubs outside the top handful of leagues, has no in-house analytics team and no enterprise sponsorship-measurement contract — is representative of the majority of the market this paper targets. (We withhold the club's name throughout this paper for commercial confidentiality; nothing in the results depends on the club's identity.)

A commercial layer already serves the sponsor-measurement need: Nielsen's sponsorship benchmarking [1], GumGum/Zoomph, Relo Metrics [2], and Shikenso all offer automated or semi-automated sponsor-visibility services. These are delivered as closed, hosted platforms: a club sends its footage to a third party and receives a report back, typically under a recurring contract, typically by integrating with an official broadcast feed. This is a reasonable model for a club with the budget, the appetite for such a contract, and a broadcast deal to plug into — but it leaves three structural costs unaddressed for everyone else: vendor lock-in and recurring fees; the requirement to hand a club's raw video and commercial sponsorship data to an external party rather than keeping it in-house; and, for the many clubs with no official broadcast coverage at all, a dependency on broadcast access that simply does not exist for them regardless of budget. Our framework instead is built around *self-service ingestion*: a club uploads whatever match video it is able to capture itself — a fixed pitch-side camera, a livestream recording, footage from a coach's or media officer's own filming — and the system produces exposure and EMV reports from that, with no dependency on an official broadcast feed or a vendor integration. We target clubs for whom these costs — not detection accuracy — are the binding constraint. To be precise about what we mean by "accessible": self-hosting removes cost and vendor lock-in, but it does not remove the need for some technical capacity to operate the system; our claim is narrower than "any club, zero technical skill" and is better stated as "clubs with modest technical capacity but no analytics budget or vendor contract."

Even with a working, open-source logo detector, two problems remain unresolved by the closest published system, ExposureEngine [3] (§II). First, the same sponsor frequently appears on both competing teams' kits, on perimeter advertising boards, or on match officials, so raw detection systematically overcounts a sponsor's exposure unless ownership is resolved per detected instance. Second, sponsors are priced by placement in practice — a chest-centre logo is worth more than one on a sock — yet no published system we are aware of attributes exposure, or price, at that granularity.

Contributions.

- **C1.** A formulation and training-free solution for *team attribution*: resolving which team owns a detected logo instance without training a separate model per opponent, using a reference-based bootstrap, appearance fusion, and track-level temporal voting.
- **C2.** A three-tier, zero-manual-setup reference bootstrap for team attribution (manual override → kit-anchor auto-clustering → luminance heuristic), so the system requires no per-match human configuration.
- **C3.** *Placement-level pricing*: attributing exposure, and therefore price, to eighteen individual kit slots via pose estimation — a granularity absent from both the manual AVE/EMV literature and recent AI-driven valuation systems (§II).
- **C4.** An open reference architecture assembled entirely from open-source models and libraries, built around *self-service video ingestion* rather than a broadcast-feed integration, with infrastructure abstracted behind swappable interfaces so that a club can start at near-zero infrastructure cost, with no dependency on having a broadcast deal, and scale only if and when needed.
- **C5.** Production-grade design and calibration lessons from a real deployment: a two-pass sampled/full-fps detection trade-off, graceful degradation under partial failure, and two specific calibration corrections (visibility threshold, train/test leakage) that changed our reported numbers substantially.

Of these, C1 and C3 are the primary technical claims; C2, C4, and C5 are supporting system contributions that make C1 and C3 practical to deploy.

II. Related Work

A. Logo detection

Logo detection is a comparatively mature problem [4]; we build on a standard fine-tuned open-weight detector rather than proposing a new detection architecture, and treat detection accuracy as a means to an end rather than as this paper's contribution.

B. Team classification and jersey-based re-identification

Assigning a detected player to a team from jersey appearance is an active area in its own right, including joint re-identification, team-affiliation, and role classification for sports tracking [5], jersey-number recognition under uncertainty [6], and earlier work combining spatial constellation features with jersey-number recognition for player identification [7]; see also the broader computer-vision- in-sports survey [8]. Most of this literature performs per-frame colour or embedding clustering for team assignment. We build directly on this and add a track-level temporal voting scheme with hysteresis, together with a zero-manual-setup reference bootstrap — the differentiation is in *how* team identity is made stable and cheap to configure per match, not in team classification itself.

C. Ad-board localization in broadcast video

A separate, older line of work localizes static advertising surfaces in broadcast video in order to insert or replace advertising content: implanting virtual advertisements into broadcast soccer video [9], automated billboard insertion [10], and billboard-advertisement detection [11]. This literature solves a related but different problem — inserting or replacing an ad on a located surface — whereas we measure the exposure of sponsors that are already, physically, on screen.

D. Automated sponsor-visibility systems

The closest published system is ExposureEngine [3], which introduces an oriented-bounding-box detector for sponsor logos and a visibility/exposure analytics pipeline for soccer broadcasts, reporting mAP@0.5 of 0.859 on a purpose-built Swedish top-flight soccer dataset drawn from official broadcast coverage. As far as its published description shows, it does not address logos shared between opposing teams or officials, does not attribute exposure below brand level, does not discuss deployment cost, and — like the commercial vendors below — assumes access to an official broadcast feed rather than club-captured video. This last point matters in practice: most clubs outside the top leagues have no broadcast coverage to draw on at all, independent of budget. Commercial vendors (Nielsen [1], GumGum/Zoomph, Relo Metrics [2], Shikenso) confirm that the underlying problem is commercially important, but their methods are closed, unpublished, and not independently reproducible.

E. Existing pricing and valuation models — manual and AI-driven

The industry-standard manual approach is Advertising/Earned Media Value Equivalency (AVE/EMV): convert the size, duration, and placement of exposure into the cost of an equivalent slot of paid advertising, typically via a hand-set placement multiplier. Our own Tier-3 pricing formula (§IV-C) is drawn from exactly this family of formulas, as used in prior industry and academic practice, and we cite it directly here rather than treating it as generic background. The AVE methodology is also the target of sustained, public criticism from industry bodies — AMEC, IPR, PRSA, PRCA, and ICCO have each issued position statements rejecting AVE as unreliable [12], citing arbitrary multipliers, no signal for content or audience sentiment, and treating every viewer as identical regardless of placement or audience quality. Our placement-level pricing (C3) makes one specific weakness of this criticism — the arbitrariness of a single, flat multiplier

— more principled, by grounding a portion of the multiplier in measured, per-slot exposure share; it does *not* address the deeper criticism that AVE-style metrics ignore audience quality, sentiment, or brand recall, and we state this as an explicit scope limitation (§VI) rather than imply the pricing problem is solved.

On the AI-driven side, recent systems apply machine learning directly to sponsorship valuation: SSPAIN.ai [13], a survival-analysis model trained on more than 5,800 historical sponsorships to forecast sponsorship renewal probability and duration, and Relo Metrics' [2] combined computer-vision-and-machine-learning pipeline for automated media value and dynamic pricing. Both are methodologically related to more general machine-learning-based dynamic pricing, e.g. in e-commerce [14], which reacts to real-time demand signals rather than sports-specific placement data. None of this AI-driven work, as far as published material shows, prices exposure below the level of a brand's aggregate on-screen time, and none discusses low-cost or self-hosted deployment.

F. Summary of the gap

No published work we are aware of simultaneously addresses (a) team/opponent-overlap attribution, (b) placement-level pricing, in either the manual or AI-driven valuation literature, and (c) deployment accessibility for clubs without an enterprise budget or an in-house ML team. This is the gap this paper targets.

III. System Architecture

The system is organized as a single orchestrated job per uploaded video (Fig. 1). Two design choices are central to the accessibility claim. First, every optional pipeline stage degrades gracefully: if a stage's dependency is unavailable (e.g. no team reference could be bootstrapped), the stage is skipped and logged rather than failing the job, so a club always gets a usable, if partial, result. Second, infrastructure is abstracted behind swappable interfaces: the default configuration uses local file storage, an embedded database, and an in-process job queue, with an optional, opt-in path to object storage, a managed database, and a distributed queue for clubs whose volume justifies it. A deployment therefore starts at near-zero infrastructure cost and only incurs additional cost when actually needed.

The pipeline also separates two passes over the video at different sampling rates: a sparsely sampled pass drives the exposure/EMV numbers (accurate enough for aggregate timing at a fraction of the compute cost), and a full-frame-rate pass produces a smooth annotated preview for human review. This trade-off is explicit and reported (§V) rather than hidden inside a single, uniformly-sampled detection pass.

IV. Method

A. Detection and annotation

The logo detector is a YOLO26m model fine-tuned at 1280-pixel input on manually annotated frames drawn from the same kind of video the framework is meant to run on in deployment — video a club captures and uploads itself, not necessarily an official broadcast feed, and therefore potentially more variable in camera stability, framing, and production quality than the broadcast-sourced footage used to train and evaluate prior published detectors. To reduce annotation cost, we use a model-assisted labelling loop: a small hand-labelled seed set (on the order of tens of images per class) trains an early model, which then proposes boxes on new frames for a human to accept, correct, or reject, before a final model is trained on the combined set. This does not eliminate manual annotation, and we treat the remaining annotation cost explicitly as the main residual barrier to full accessibility (§VII), rather than claiming an annotation-free pipeline in this paper. Per-frame detection sits behind a swappable backend interface, so a transformer-based detector (RF-DETR) can replace the YOLO backend without touching the rest of the pipeline; all detection numbers reported in this paper use the YOLO26m backend. In deployment, detections below a confidence floor of 0.25 are discarded, but confidence is otherwise used as a continuous weight in the visibility score (§IV-C) rather than as a gate — a deliberately recall-first choice: in our deployment data, detections with confidence below 0.4 account for 29% of pose-attributed detections but only 9.5% of quality-weighted exposure, so genuine small or motion-blurred kit logos are retained at little cost to the aggregate, and spurious ones are further suppressed by the minimum-duration rule of §IV-C.

B. Team attribution

Detected persons are tracked across frames; for each track, a jersey crop is classified by fusing a colour histogram with a general-purpose vision embedding, and classifications are accumulated per track with hysteresis, so that a single blurred or occluded frame cannot flip an already-confident team label. Reference examples for each kit are obtained automatically rather than by manual setup: from official kit images when available, and otherwise by clustering the video's own early frames and selecting the cluster closest to a simple luminance heuristic (e.g. a known dark alternate kit). Each detected logo instance is then attributed to the team of its nearest tracked person. When the evidence for opponent ownership is insufficient (a track has not yet accumulated enough confident votes), the instance is *kept* rather than dropped — an explicit, revenue-safe design choice: an under-confident drop would silently understate a paying sponsor's measured exposure, which we consider a worse failure mode than a small number of false positives that a human reviewer can catch.

C. Visibility, exposure, and EMV pricing

Exposure and pricing are computed in three tiers, adapted from the manual AVE/EMV family discussed in §II-E rather than proposed as a new pricing theory: (1) a per-frame *visibility* score combining detection size, on-screen position, and detector confidence; (2) temporal aggregation of visibility into per-brand *exposure* segments, discarding very short, likely spurious detections; and (3) a conversion to *estimated media value* using an audience size, a cost-per-thousand (CPM)

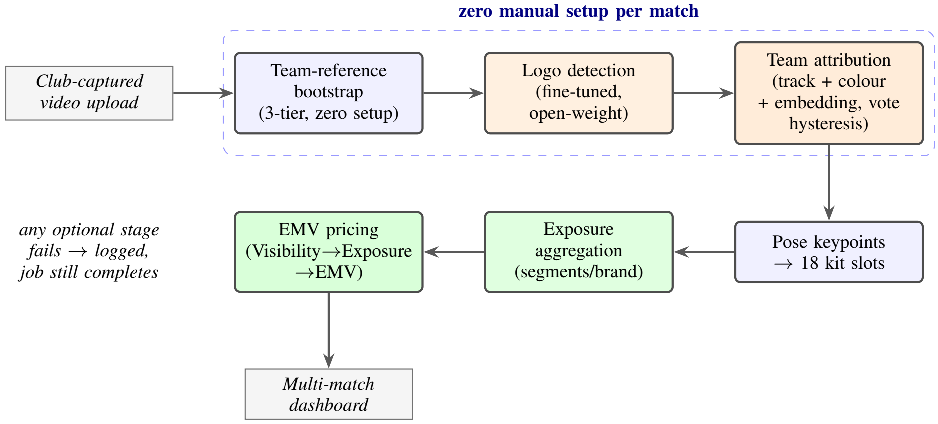

Fig. 1. **End-to-end pipeline.** An orchestrated job runs frame sampling, a zero-manual-setup team-reference bootstrap, logo detection, team attribution, pose-based kit-slot assignment, exposure aggregation, and EMV pricing, feeding a multi-match dashboard. Every optional stage degrades gracefully on failure rather than aborting the job, and infrastructure (storage, database, job queue) sits behind swappable interfaces so a deployment can start at near-zero cost.

baseline, and a placement multiplier supplied per deployment (the implementation additionally supports sponsor-category and prime-time multipliers, both disabled in the deployment reported here). A specific calibration lesson from deployment: the visibility floor of 0.1 suggested in the pricing methodology we adapt, applied as published, discards nearly all real, small, off-centre sponsor logos typical of kit sponsorship — under our size term such logos score in the 0.03–0.08 range — so we use an empirically set floor of 0.02 (**report the sensitivity of exposure-seconds to this visibility floor and to the detector confidence floor of 0.25 jointly**).

D. Placement-level pricing

Each attributed detection is additionally assigned to one of eighteen kit slots (e.g. chest-centre, sleeve, sock) via pose-estimation keypoints, so that exposure — and hence the placement multiplier applied in Tier 3 — can be reported and priced per slot rather than only per brand (the measured per-slot distribution from our deployment is shown in Fig. 3). This directly targets the "arbitrary multiplier" criticism of AVE-style pricing (§II-E) by grounding part of the multiplier in measured, slot-specific exposure share, while leaving audience-quality and sentiment signals as future work (§VI).

V. Experiments

A. Setup

We evaluate on match video self-ingested by our partner club (identity withheld) — video the club captured and uploaded itself, not an official broadcast feed. The annotated detection dataset covers seventeen sponsor classes and is severely imbalanced, from roughly 200 instances for the most frequent kit sponsor down to single digits for the rarest — a property of real kit sponsorship, not of our sampling, since a chest sponsor simply appears in almost every frame while a shorts-leg sponsor does not. Detector training and evaluation use a clip-disjoint split: all frames from the same clip or match are kept entirely within one split. This choice matters more than any architectural one we made: because analytics frames are sampled at 2 fps, neighbouring frames are near-duplicates, and a naive random-frame split leaks them across train and validation. The same YOLO26m detector scores mAP@0.5 = 0.86 under a random-frame split but 0.70 under the clip-disjoint split of the same source data (Table I) — the higher number reflects leakage, not a better model, and we report the pair explicitly because published results in this domain rarely state which split strategy was used.

TABLE I
Effect of the train/validation split policy on measured detection accuracy (fine-tuned YOLO26m, best validation epoch). The random-frame figure is inflated by near-duplicate frames leaking across the split, not by a better model.

Split policy	mAP@0.5	mAP@[.5:.95]
Random-frame	0.86	0.57
Clip-disjoint, same data	0.70	0.42
Clip-disjoint, extended data	**0.75**	**0.44**

B. Detection accuracy

On the initial annotated set, the clip-disjoint validation score is mAP@0.5 = 0.70 (mAP@[.5:.95] = 0.42); after extending the dataset with additional annotated clips under the same clip-aware split policy, the current detector reaches mAP@0.5 = 0.75 (mAP@[.5:.95] = 0.44, precision 0.65, recall 0.74). Rare-class AP is unreliable at these instance counts and the validation split intentionally holds no instances of the rarest classes, so aggregate mAP should be read together with the per-class breakdown. **Report per-class average precision and a confusion matrix on a held-out clip-disjoint test split (current numbers are validation-split figures used for model selection).**

C. Team attribution

Report precision/recall of team attribution on a labelled clip sample spanning more than one match and lighting

condition (current evidence: two case-study clips only — insufficient for a quantitative claim and must be extended before submission).

D. Team-attribution counterfactual

Across nine full-match highlight videos processed by the deployed pipeline, team attribution flagged and excluded 44% of raw logo detections (11,161 of 25,153) as owned by the opposing team or match officials, or as not attached to any tracked player (e.g. perimeter advertising boards). We did not take this aggregate at face value: we visually audited the per-track attribution overlays of all nine matches. Five show correct target/opponent assignment; three (two night-time, long-range club-feed fixtures and one fixture in which both teams wore dark kits) work overall but exhibit per-track errors in both directions; and in one — the match with the highest exclusion rate, 78% — the fused classifier systematically assigned the target club's own white kit to the opponent class under night floodlighting, despite reference features that a luminance check confirms are correct. We report that last match as a failure case (§VI), not as filtering, and exclude it from the headline figure: over the remaining eight matches, attribution excluded 41% of raw detections (9,383 of 22,863), with per-match rates of 21–50% on dark-kit fixtures and 26–72% on white-kit fixtures — the upper end driven by the noisy night-time captures above, in which many distant tracks never accumulate a confident label. Without attribution, every excluded detection would have been counted — and priced — as target-club kit exposure. These figures quantify how much raw exposure the filter removes in production, not yet its per-instance accuracy, which is the subject of the labelled-clip evaluation of the previous subsection. To isolate the value of C1 further, we additionally run the pipeline with team attribution enabled and disabled on clips where a sponsor appears on both teams' kits, and report the resulting difference in measured exposure-seconds and EMV. **Report the counterfactual over-counting avoided on shared-sponsor clips, e.g. "X% lower EMV / correctly attributed exposure-seconds with team attribution on."**

E. Exposure and EMV validation

Fig. 2 shows the per-sponsor exposure segments the pipeline recovers for one full highlight video, which is the intermediate representation all exposure and EMV figures are computed from. To validate it, we compare system-computed exposure-seconds against an independent manual timing of logo appearances for at least one full match. We report EMV as a downstream consequence of this validated exposure measurement rather than as independently verified: CPM, audience size, and placement multiplier are deployment-supplied business inputs, not measurable ground truth, so multiplying by them is arithmetic rather than something to validate against an external reference — unless an independent buyer valuation is available for direct comparison, which would be a stronger, separate claim. **Report exposure-seconds MAE against manual ground truth.**

F. Placement-level exposure in deployment

Fig. 3 reports the measured per-slot exposure distribution that placement-level pricing (C3) is built on. Two observations matter for pricing. First, the distribution is far from uniform: shoulder and chest-adjacent slots dominate attributed exposure, while sleeve, sock, and back-centre slots individually carry under 2% each — supporting the practice of pricing slots differently, but from measurement rather than convention. Second, the ordering is not the conventional price ordering: in this deployment the shorts-back slot outranks chest-centre, an artefact of how the club's own pitch-side camera views play, and precisely the kind of deployment-specific evidence a flat industry multiplier cannot anticipate.

G. Accessibility evidence

Report end-to-end processing time (video-minutes processed per wall-clock minute) on one named, commodity hardware configuration, and the approximate annotation time spent per sponsor class using the model-assisted labelling loop of §IV. These two numbers are the primary quantitative support for the accessibility claim in this paper and are prioritized for measurement before submission.

VI. Discussion and Limitations

Team attribution assumes a stable kit within a match; a mid-season kit change requires a reference refresh. Colour-and-embedding fusion for team classification has not yet been stress-tested across a wide range of lighting conditions, and this is not hypothetical: in one night fixture in our deployment, the classifier systematically assigned the target club's own white kit to the opponent class even though the stored reference features were correct (§V-D) — silently deleting most of that match's genuine sponsor exposure. Because the failure direction is revenue-critical, a deployment should monitor the per-match exclusion rate and treat an outlier (here, 78% against a fleet median near 40%) as a signal for human review rather than as a valid measurement. Because the framework targets self-captured video rather than official broadcast feeds, it must also tolerate a wider range of camera stability, framing, and production quality than prior systems evaluated only on broadcast-sourced footage; we have not yet characterized detection or team-attribution robustness across this range of capture sources (e.g. a fixed pitch-side camera vs. a hand-held recording), and flag this as an open question rather than a validated property. The sparse sampling rate used for the analytics pass trades exposure-time precision for processing cost; we report the resulting worst-case timing error explicitly rather than treat sampling as free. Manual annotation, while reduced by the model-assisted loop, remains a real, non-eliminated cost when onboarding a new sponsor class, and is the main residual barrier to the accessibility claim in this paper (§VII). As discussed in §II-E, our placement-level pricing addresses one specific, well-documented weakness of AVE-style valuation (arbitrary multipliers) but not the broader critique that such metrics do not capture audience quality, sentiment, or brand-recall impact; we do not claim to have solved sponsorship

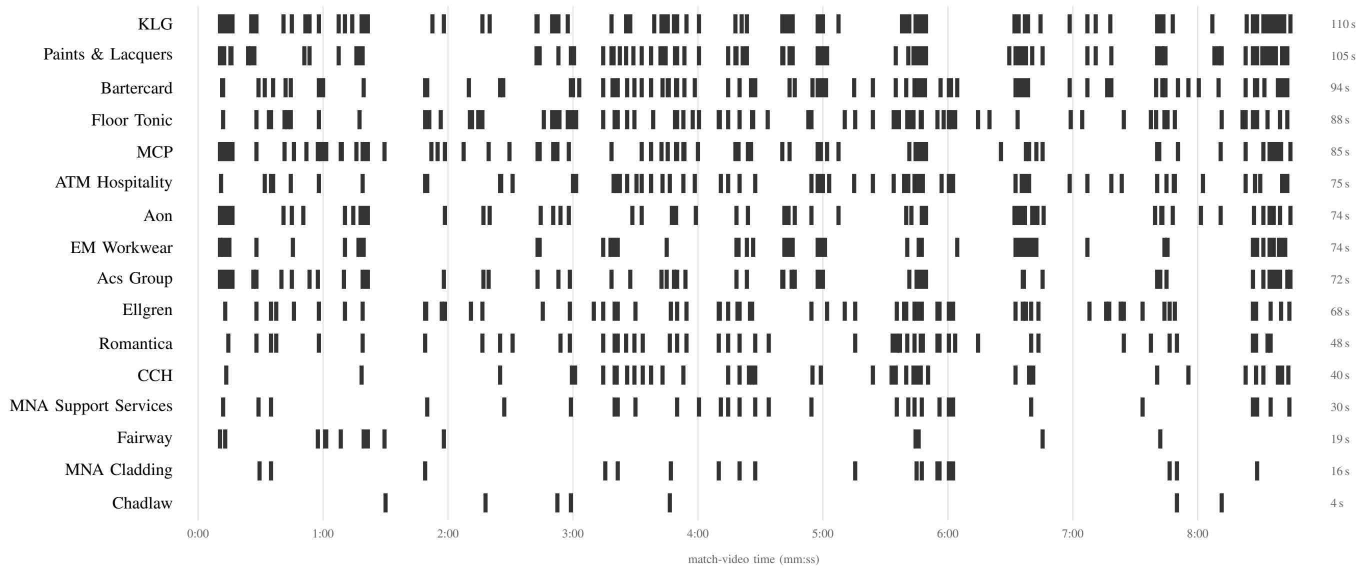

Fig. 2. **Per-sponsor exposure timeline produced by the deployed pipeline** for one club-captured highlight video (8.9 min, black-kit fixture, 2 fps analytics pass, team attribution enabled; this match's attribution overlay was visually audited and assigns target/opponent correctly, §V-D). Each bar marks an interval in which the sponsor was detected on the target team's kit after attribution; sponsors are sorted by total on-screen time (right column). Very short intervals are widened to a minimum drawable width for legibility; totals are computed from the underlying segments, not from the drawing.

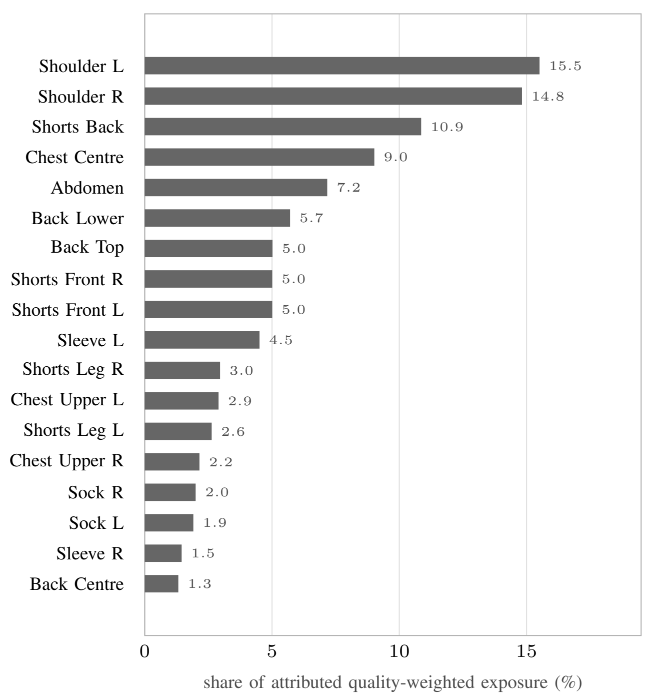

Fig. 3. **Where kit exposure actually accrues: share of quality-weighted exposure per kit slot**, aggregated over eight full-highlight analyses (§V-D; the ninth match is excluded after the attribution audit described there, since its kept detections are not trustworthy). Percentages are weighted by each match's attributed exposure and cover only detections the pose stage could assign to a slot. The distribution is strongly non-uniform — the two shoulder slots alone carry ∼30% of attributed exposure, an order of magnitude more than a sock slot — which is exactly the structure a single flat placement multiplier cannot express.

valuation, only to have made one component of it less arbitrary and more transparent.

Data and ethics statement. The framework is designed around video that a club captures and uploads itself, which removes the third-party broadcaster copyright concerns that would apply to processing official broadcast feeds; the club retains ownership of its own footage. The system nonetheless processes images of real players, so consent/data-handling practice still matters. **state precisely what permission or policy governs use of a club's own match footage for this research, and for any figures published in the paper.** Players are tracked only as anonymous instances for the purpose of kit-slot attribution, not for personal identification; the system's output is brand and media-value analytics, not player-performance or biometric data.

VII. Future Work

The main remaining barrier to full accessibility is the manual annotation effort required per sponsor class. We are separately exploring an annotation-free direction that exploits a structural property of team sports: a club wears one fixed kit for an entire season, and advertising boards are physically static, so each distinct logo-bearing surface could, in principle, be identified once — at its clearest occurrence in the available footage — and propagated to every other occurrence by tracking and geometry rather than by repeated visual classification. Early internal results on this direction are encouraging but preliminary, and are explicitly not part of the claims made in this paper. Other directions include synthetic data generation for rare viewing conditions, and a more detailed, audience-quality-aware model for converting measured exposure into media value, directly addressing the deeper AVE critique noted in §VI.

VIII. Conclusion

We present an open, self-hosted framework for sponsor exposure and media-value measurement that addresses two problems left open by prior published work — attribution of logos shared between competing teams, and pricing at the level of individual kit placements — while remaining deployable without an enterprise vendor contract or an in-house machine-learning team. We position our pricing methodology explicitly against both the manual AVE/EMV tradition and recent AI-driven valuation systems, and are transparent about which part of the well-documented criticism of media-value-equivalency metrics our contribution does, and does not, address. The framework is deployed and evaluated on a real professional club, which grounds the accessibility claim in an actual deployment rather than in argument alone.

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